NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L22.4 Literature figure gallery and sentence-level retrieval

Group First Lab (proving ground)

Status Pending promotion

Period 2026-10-15 17:58 — 2026-10-18 01:26

Stage conclusion

10-17 Conclusion: A figure-caption retrieval is worth promoting into a platform tool (96% pairing accuracy, English figure search 19/20, Chinese 12/20; 5/5 conditional approval in the 10-17 review, `figs` is live; 10-18 re-evaluation passed pairing at 92% but failed English figure search at 13/20, so center-wide release is on hold pending platform fixes and re-testing on new questions, with 8 launch conditions including post-fix re-evaluation; 16–17/20 in English under the strict criterion); A2 image-to-image search (5/10) will not be pursued; B sentence-level clustering underperformed page-level retrieval on all three questions (clusters by topic rather than stance, and reference entries drown out conclusion sentences), will not be pursued, and full-corpus encoding has been stopped

Project report

Literature Figure Library and Sentence-Level Retrieval

Objective

On top of the `ftkb` full-text library (Zotero 3,149 + EndNote 1,527 + Desktop 12,491 documents, ~600,000 pages), evaluate whether two capabilities are worth promoting into platform tools: (A) a figure library (extracting figures + captions across the entire corpus for caption-text and visual-semantic retrieval); and (B) whether sentence-level indexing + clustering is clearly superior to page-level retrieval. Deliver a one-sentence conclusion for each of A and B by 10-21. Task brief (approved by Liang Hu on 10-15).

Current Status

10-17 13:0x: **A/B conclusions reached** (ahead of 10-21): A figure-caption retrieval is worth turning into a platform tool (96% pairing accuracy, English figure search 19/20; add a Chinese translation path and single-column crop fixes before productization); A2 image-to-image search will not be pursued (5/10); B sentence-level clustering will not be pursued (underperformed page-level retrieval on all three questions for structural reasons). B full-corpus encoding was stopped at 10-17 16:18 (172/469 shards; shards and logs retained). Handing A over to the platform for integration constitutes promotion, and Liang Hu has been asked. Conclusion.

Settled Decisions

- 2026-10-15 Node established, gated on prerequisite platform update →
- 2026-10-15 Restrict to preparation (scripts + trial run + frozen question bank) while prerequisite platform update is pending; figure-finding questions authored and frozen by Lead →

Settled Decisions (Continued)

- 2026-10-15 A1 v1 pairing accuracy below threshold; revise three rule categories and rerun →

- 2026-10-15 Revised version passes pairing at 92%, fix 76% cropping completeness first, move both libraries to desktop processing →
- 2026-10-15 Desktop read-only fault; pause A1 pending reboot →
- 2026-10-15 Desktop not rebooted; detour by writing to the Windows drive →
- 2026-10-16 Desktop rebooted and recovered with data intact; switch back to writing on root drive →
- 2026-10-16 A1 preparation complete, waiting only on prerequisite platform update; ask Liang Hu whether to launch early →
- 2026-10-16 Prerequisite platform update went live, A1 full run launched →
- 2026-10-16 A1 decision: pairing 96% and English 19/20 passed, Chinese 12/20 failed; dispatched A2 →
- 2026-10-17 A2 did not pass the bar; image-to-image search will not be pursued →
- 2026-10-17 A/B conclusion: pursue A, do not pursue A2 or B →

Criteria (Pre-Specified in the Brief)

Item	Criterion
A1 Caption Index	≥90% pairing accuracy on 50 random figure–caption pairs; top-5 hit on ≥14/20 figure-finding questions
A2 Image Embeddings	Given 10 of our own simulation figures to find reference figures in the literature, top-5 contains a relevant figure on ≥6/10 queries (executed only if A1 passes)
B Sentence-Level + Clustering	Across 3 questions (climate-change wind-speed trend + 2 EVA questions), compare representative sentences + source tables per cluster against `pages` page-level results; Lead judges whether it is clearly better

Trials and Results

Date	What Was Done	One-Line Result
10-15	EVA line provided 2 mixed-climate questions for B	Impact of separate vs. combined fitting on long-return-period design wind speeds; misclassification and missed-observation bias

Date	What Was Done	One-Line Result
10-15	A1 v1 trial run on 400 documents (`ideas`) + Lead judgment on 50 pairs	Pipeline works end-to-end; strict caption match 19/40, relaxed 35/40, 10 table titles, 7 false captions, 16 full-page crops; decoding failed on 120/400
10-17	B sentence-level fast path on three questions (`ideas`) + Lead verdict	Clustering silhouette 0.07–0.11, representative sentences mostly reference entries; all three questions underperformed page-level top 10
10-17	A2 image embeddings (`ideas`, CLIP ViT-B/32) + Lead verdict	387k figures encoded in 2.3 h; top-5 across 10 queries: image-to-image 5/10, text-to-image 3/10, below the ≥6/10 bar
10-16	A1 full-scale evaluation (Lead)	50-pair matching 48/50=96%, crop completeness 34/50; 20 queries top-5: English 19/20, Chinese 12/20
10-16	A1 full-scale figure extraction (`ideas`, desktop 12 cores)	17,167/17,167 items processed, 386,894 figures; 3,998 items had partial skipped figures, 0 whole-document failures; compact index 116 MB (<200 MB), thumbnails ~11 GB; caption vector encoding running
10-16	A1 post-reboot continuation (`ideas`) + Lead judgment on v3 20 pairs	All 4,676 PDFs staged; Zotero v3 rerun finished; 20-pair check scored 20/20 pairing and 18/20 (90%) cropping completeness
10-15	A1 pre-full-run preparation (`ideas`, v3)	v3 cropping rules implemented; Desktop 200 + EndNote 50 re-rendered; 2,314/4,676 PDFs staged across the two libraries; interrupted at 23:1x when desktop turned read-only
10-15	A1 revised rerun on 400 documents (`ideas`) + Lead judgment on new 50 pairs	Pairing 46/50=92% (passed); complete crop 38/50; 1 false caption, 0 table titles, 0 mid-run aborts across 400 documents; kind split: raster 36%[local path omitted] 22%/`page_region` 42%
10-15	Lead authored and froze 20 figure-finding questions	Covers downbursts, Cp, typhoon profiles, gust factors, spectra, bluff bodies, response control, extreme-value standards, etc.

Parameter Source Table

Parameter	Value	Source or "Assumed"	Where Checked
Thumbnail size	Long edge ≤512 px JPEG	Assumed (sufficient to read plot types and axes while keeping storage manageable)	No literature basis
Caption block detection	Starts with Fig.[local path omitted] "Tu" + number, not followed by predicate verbs like shows/gives, short block, adjacent to figure	Assumed (inductively refined from the 50-pair error analysis)	In-line trial runs
Full-corpus sentence count	~15 million sentences	Task brief estimate (600,000 pages × ~25 sentences)	Pending empirical update from B

Open Items and Next Steps

- Prerequisite platform update go-live time → Lead follow-up check at 10-16 08:30; once live and desktop OCR is complete, notify `ideas` to launch full figure extraction (≤16 cores).
- A1 trial report and 50-pair sample → Lead inspects images to judge pairing rate; after full run, Lead runs the 20 questions to score hit rate.
- B dispatched to `ideas` (10-16 12:5x, desktop ≤8 cores, due 10-20) → Lead compares against page-level baseline to judge whether it is "clearly better"; share results with the EVA line.
- Dispatch A2 only after A1 passes; Lead selects 10 in-house simulation figures from public figures across our lines and the CFD line.

Literature Basis

- Jian Zhang and Jia Shao, *Restructuring Higher Education Curricula, Textbooks, and Pedagogy with Small Language Models* (materials provided by Liang Hu): source of the sentence-level corpus + unsupervised clustering methodology and scripts; figure library retrieval originates from the accompanying slide deck (no code included in the package).
- No other literature basis; based on in-line data.

Notes from Liang Hu

10-17 17:3x Review and Platform Handoff

- Review: 5/5 conditional approval → handed over to the platform; see the review verdict and the 8 launch conditions.
- Correction: `fig-06` and `fig-08` in English were revised to *X* (the top 4 results for `fig-06` are downburst papers); under the strict criterion the English score is 16–17/20, and 19/20 is superseded.
- Next steps: once the platform fixes the index, arrange a re-evaluation on 20 new questions authored and scored by someone other than the Lead (≥14/20 in both Chinese and English) + re-sample 50 pairs.
- 10-17 platform reply: item opened and will be accepted against the 8 conditions in the verdict; restricted to the Lab only until Condition 6 re-evaluation passes; scheduling will be forwarded after assessment in the platform session.
- 10-18 Condition 6 re-evaluation in progress: (1) 50 pairs judged by Lead, with pairing at 46/50=92% (passed) and cropping completeness at 42/50=84%; (2) 20 new questions independently authored by Reviewer E (given only the tool description without seeing the index), to be frozen upon return and handed to `ideas` to run `figs` and score strictly against caption + bibliographic record + thumbnail.
- 10-18 00:36 The 20 new questions have been received and frozen (authored by Reviewer E, sha256 verified) and handed to `ideas` to run and score (≥14/20 in both Chinese and English).
- 10-18 00:5x In the first pass, `ideas` declined to inspect thumbnails per line 29 of its spec and returned "0/20 unconfirmed" (not a performance score). Updated division of labor: `ideas` assigns A/B/C from caption + bibliographic metadata and compiles the image-inspection queue, while the Lead inspects each image for final confirmation (Lead is already inspecting the 50 pairs under the same criterion).
- 10-18 Condition 6 re-evaluation completed: pairing passed at 46/50=92%; on the 20 new questions, Chinese scored 15/20 and English scored 13/20, so English fell short of the pre-specified criterion → not opened center-wide for now, remaining in trial use within the Lab; the issue list has been handed over to the platform for fixes, to be re-tested on another new question set (decision).

Key figures